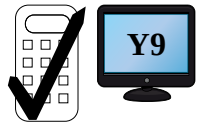


Shape Worded Questions



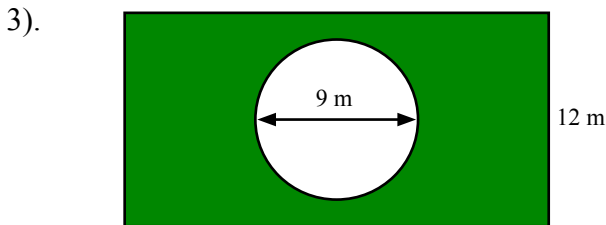
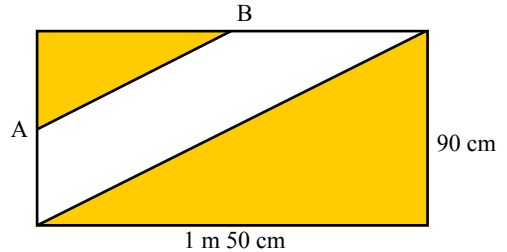
Take $\pi = 3.14$. Leave answers to an appropriate degree of accuracy.

- 1).
 - a). Write the formulae for the area and circumference of a circle.
 - b). A circle has a radius of 8 cm. Find its area and circumference.
 - c). A circle has an area of 350 cm^2 . Find its radius, then its circumference.

- 2). A rectangular flag has two colours. The measurements are as shown in the diagram.

The line AB is drawn from the midpoints of either side.

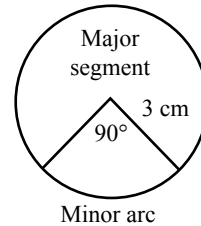
- a). Calculate the area of the flag.
- b). Calculate the shaded area of the flag.
- c). Calculate the white area of the flag.
- d). The material to make the flag costs £8.20 per m^2 . How much will the material for the flag cost?



A rectangular garden has a fruit section in the shape of a circle, with a diameter of 9 metres. Find

- a). the area of the circular fruit section,
- b). the shaded area.

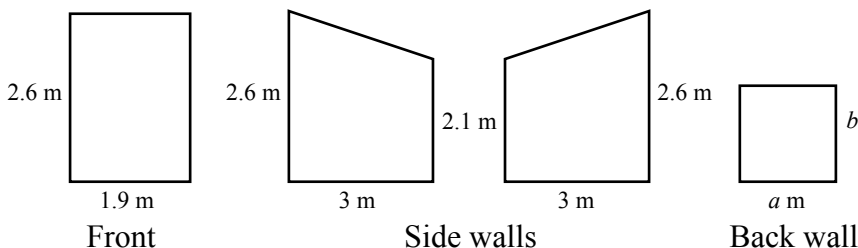
- 4). In the diagram the radius of the circle is 3 cm.
 - a). Find the area of
 - i). the minor segment,
 - ii). the major segment.
 - b). Find the
 - i). minor arc length,
 - ii). major arc length.



- 5). The Millenium Wheel (The London Eye) was the world's tallest observation wheel when it was built. It stands 135 m high. Take the height as its diameter.
 - a). What is its circumference?
 - b). It has 32 pods on its circumference. It takes a pod 30 minutes to complete one revolution. At what speed is the pod travelling in km/h?



- 6). A shed is made up of the following shapes. The floor is a rectangle



- a). Work out the area of the front.
- b). What is the width, a , of the back wall?
- c). What is the height, b , of the back wall?
- d). Work out the area of the back wall.

- e). To find the area of the side wall it can be split into a triangle and a rectangle.
 - i). Find the area of the rectangle.
 - ii). Find the area of the triangle.
 - iii). Find the area of the side wall.
- f). Work out the total area of the four walls.

